

was a real pain; it's done by downloading a utility from Hitachi's Web site that needs to be installed on a floppy (which becomes bootable). Alternatively, an ISO image file (in case you don't have a floppy drive handy, like us), can also be used.

The problem was, our downloaded ISO turned out to be corrupt. In fact, it took three downloads and a writer than could burn at incredibly low speeds to get the ISO working. All's well that ends well and the performance difference was worth the effort for us. A relatively new user is likely to have a lot of problems getting this right. Kudos to the other three manufacturers for designing a simple workaround.

Some of the SATA drives from WD and Seagate featured a 16 MB cache, which makes these drives more capable—at least theoretically—at dealing with large file transfers. Among the others only the Hitachi 500 GB SATA had a 16 MB buffer. All others had 8 MB buffers, which we feel is more than sufficient for most usage needs.

Most of the newer SATA drives do not have the older 4-pin power connectors, rather they have the newer SATA connectors. Backward compatibility can be a problem with older SMPS's that don't have these connectors. Adapters are thankfully available in abundance, in case your power supply doesn't provide these.

Seagate has the distinction of having sent in the largest drive in this shootout, their latest

7200.10 series Barracuda, with a capacity of 750 GB. This drive features Perpendicular Recording technology, which allows cramming a previously unheard-of 188 GB on each platter (the drive uses four platters). This drive performed better than all the other drives in many of the tests, but whether this is due to the technology used, or a variety of other factors, it's just too early to tell. Judgment withheld until more drives built around Perpendicular Recording technology emerge, at which time we can do a comparison, pitting a good number of such drives against older drives.

Performance HD Tach 3.0.1.0

We fired up the HD Tach 8 MB file tests first (followed by the 32 MB ones). While the burst rate scores aren't really significant, we noted that three of the Seagate SATA drives topped the burst transfer rate test. This shows that the Seagate drives are making the most of the additional theoretical bandwidth that SATA 2 provides.

Only four drives cracked the magic 60+ MBps in the average read tests: the massive 750 GB Barracuda scored 67 MBps, followed by the 400 and 500 GB WD SATA siblings with scores of 64 and 62.4 respectively. Among the 250 GB drives, the Samsung SP2504C was the topper, with another 60+ score. The scores in both the HD Tach tests (that is, the 8 MB and 32 MB file samples), were nearly identical. A couple of drives from Samsung

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Hitachi 500 GB
HD5725050KLAT80

How We Tested

For our internal drive shootout, we used an AM2-based system, powered by a 2.6 GHz 5000+ processor, 2 GB of DDR2 800 MHz memory in dual-channel configuration, and an XFX 7900GTX graphics card. The Hitachi 7K 250 GB SATA 2 HDD was used as our test bench hard disk. The hard disks to be tested were kept clean from any unnecessary software installs, except for the test files. All drivers and necessary software were installed on the test hard drive.

All the hard disks were set as master drives (except the SATA drives, which have individual channels). SATA 2 drives were run at the maximum transfer mode, that is, 3 Gbps, while the SATA 1 drives were run at 1.5 Gbps. Two partitions each of 50 GB were made on each of our test drives, except for the smaller drives, where the primary partition was 30 GB, with the remaining size devoted to the second partition.

We installed Windows XP Professional with Service Pack 2, with all the latest drivers installed.

Our synthetic benchmarks consisted of HD Tach 3.0.1.0 and SiSoft Sandra 2007 Engineer. HD Tach is a very trusted low-level benchmark that needs a completely fresh, that is, unformatted, unpartitioned hard drive to run. The scores it returns are very close to the actual raw speed that a hard drive can attain, simply because it doesn't access the hard disk through Windows and various other software layers, which slow down the drive and adversely affect scores. Because of this uniqueness of HD Tach, we fired it up first.

We ran both 8 MB and 32 MB file sample tests. The Burst Speed figure is a good indicator of what the bus is capable of, but it loses significance as a real-world figure. This is because it represents a "burst" or instantaneous value. The Average Read and Write figures reflect the real-world performance the drive is capable of, as these are the sustainable data transfer rates.

SiSoft Sandra 2007 is another very popular benchmark that tests various facets of a drive's performance. SiSoft Sandra gives a

Drive Index figure—a combined average of the Read and Write tests, which are divided into buffered, random, and sequential read and write tests. Besides these tests, SiSoft Sandra also reports the access time.

For our real-world tests, we used a single 4 GB .rar file to test sequential read and write performance. For random read/write operations we used a combination of more than 25 different file types. The files were also purposely placed in a multitude of folders, and the size of the files was also greatly varied. There was a Word document occupying exactly 25 KB of disk space, while an AVI file was 400 MB in size! The total file size was 4 GB, the same as that of the single .rar file. The variance in size is a good way to test how a drive performs during real-life data transfers, involving lots of small files, as in copying multiple MP3 files from one drive to another.

For our File Write tests, we copied from the test bed drive to the test hard drives. For the File Read tests, we simply copied the test file back to the test bed drive. Our internal file test was the most strenuous test (as the figures show), where we copied the test files from one partition on the test drive to the other partition.

We installed a fresh copy of Adobe Photoshop CS 2 on our test drives. The scratch disk was also kept on the same partition as the install. The test PSD files were also placed on the same partition, a true test of the mettle of today's drives.

Finally, for our game benchmark, we installed a fresh copy of *FarCry* with patch 1.32 installed, on the drive to be tested. The first map was loaded, and the time taken to load was noted. We used Map 1, simply because it starts with a cutscene that isn't very graphics-intensive, as opposed to other levels, which start with gameplay: this causes a slight pause, because it takes the graphics card a little longer to render the scene. We wanted the graphics card to play as small a part as possible, so we noted the time just after the starting of the cutscene cinematic.